Vector Spaces III: Sums, Intersections and Direct Sums

From the class book: Chapter 3, Section 3.5

1. Key definitions

Sum and intersection. For subspaces W_1, W_2 of V: $W_1 \cap W_2$ is a subspace, and $W_1 + W_2 = \{w_1 + w_2 : w_i \in W_i\}$ is the smallest subspace containing $W_1 \cup W_2$. The union $W_1 \cup W_2$ is a subspace only when one of them contains the other.

Direct sum. $V = W_1 \oplus W_2$ if $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$; equivalently, every $v \in V$ is **uniquely** $w_1 + w_2$. A **complement** of W is any W' with $V = W \oplus W'$.

- For several subspaces, $W_1 + \cdots + W_k$ is direct iff each W_i meets the sum of the others only in 0, iff $\dim \sum W_i = \sum \dim W_i$.
- Pairwise zero intersections are **not** enough for three or more subspaces: three distinct lines in a plane.

2. Main results

Theorem (Dimension of a sum, Grassmann's formula)

- $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$.
- With $\dim V = n$ and $\dim W_1 = m_1 \geq \dim W_2 = m_2$: $\max(0, m_1 + m_2 - n) \leq \dim(W_1 \cap W_2) \leq m_2$.
- And $m_1 \leq \dim(W_1 + W_2) \leq \min(m_1 + m_2,\ n)$.
- In particular, if $m_1 + m_2 > n$ then $W_1 \cap W_2 \neq \{0\}$: two planes through 0 in $\mathbb{R}^3$ always share a line.

Important results: direct sums

- If $V = W_1 \oplus W_2$, then $\dim V = \dim W_1 + \dim W_2$, and the union of bases of W_1 and W_2 is a basis of V.
- If $\dim W_1 + \dim W_2 = \dim V$, then any one of $W_1 + W_2 = V$ and $W_1 \cap W_2 = 0$ implies the other.
- Every subspace has a complement (extend a basis of W to one of V); complements are **not** unique. A line is a complement of a plane in $\mathbb{R}^3$ iff it does not lie in the plane.

Decomposition	Formula
$M_n(\mathbb{R}) = \{\text{symmetric}\} \oplus \{\text{skew-symmetric}\}$	$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$
$M_n(\mathbb{C}) = \{\text{Hermitian}\} \oplus \{\text{skew-Hermitian}\}$ (over $\mathbb{R}$)	$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*)$
Real functions $= \{\text{even}\} \oplus \{\text{odd}\}$	$f = \frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$; e.g. $e^x = \cosh x + \sinh x$
$M_n = \{\text{upper triangular}\} \oplus \{\text{strictly lower}\}$	split along the diagonal (upper + lower is **not** direct)
$M_n(F) = \{\operatorname{tr} = 0\} \oplus \{cI\}$	$A = (A - \frac{\operatorname{tr} A}{n} I) + \frac{\operatorname{tr} A}{n} I$; needs $n \neq 0$ in F

3. Methods

Method: $\dim(W_1 + W_2)$ and $\dim(W_1 \cap W_2)$.

- Spanning sets given: put all spanning vectors of W_1 and W_2 as rows of one matrix; its rank is $\dim(W_1 + W_2)$. Then get $\dim(W_1 \cap W_2)$ from Grassmann.
- Equations given: stack the equations of W_1 and W_2; $\dim(W_1 \cap W_2) = n - \operatorname{rank}$ of the stacked system. Then get $\dim(W_1 + W_2)$ from Grassmann.
- Coordinate subspaces ($x_i = 0$ for $i \in A$, resp. $i \in B$): the intersection kills $A \cup B$, the sum kills only $A \cap B$.

Method: is the sum direct?

- Check $\dim W_1 + \dim W_2 = \dim(W_1 + W_2)$, or find a non-zero common vector.
- To find a common vector, solve $a_1 u_1 + \cdots = b_1 v_1 + \ldots$ (a homogeneous system in the coefficients).

4. Exam shortcuts and common traps

- Only $\dim S$, $\dim T$, $\dim V$ given? The least possible $\dim(S \cap T)$ is $\dim S + \dim T - \dim V$ (or 0), and the greatest is $\min(\dim S, \dim T)$; all values in between occur.
- $\dim(\text{symmetric} \cap \text{upper triangular}) = n$ (the diagonal), so symmetric + upper triangular is all of M_n.
- "$\dim(W_1 \cap W_2)$" and "number of coordinates forced to 0" are different numbers. In $\mathbb{R}^{100}$, $\{x_i = 0 : 4 \mid i\} \cap \{x_i = 0 : 5 \mid i\}$ has dimension $100 - (25 + 20 - 5) = 60$.
- **Traps:** $W_1 \cup W_2$ is almost never a subspace; complements are not unique; three subspaces with pairwise zero intersections need not form a direct sum; over $\mathbb{F}_p$ with $p \mid n$, the scalar matrices have trace 0.

5. Solved examples

Example 1. $S, T \subseteq \mathbb{R}^{24}$ with $\dim S = 19$, $\dim T = 17$. Find the least and greatest possible $\dim(S \cap T)$.

Solution

- $\dim(S \cap T) = 36 - \dim(S + T) \geq 36 - 24 = 12$.
- $\dim(S \cap T) \leq 17$, with equality when $T \subseteq S$.
- All values $12, \ldots, 17$ occur (coordinate subspaces).

Example 2. In $\mathbb{R}^4$, $W_1 = \operatorname{span}\{(1,0,1,0), (0,1,0,1)\}$, $W_2 = \operatorname{span}\{(1,1,0,0), (0,0,1,1)\}$. Find $\dim(W_1 + W_2)$ and $W_1 \cap W_2$.

Solution

- The four vectors have rank 3: $(1,0,1,0) + (0,1,0,1) = (1,1,0,0) + (0,0,1,1)$.
- So $\dim(W_1 + W_2) = 3$ and $\dim(W_1 \cap W_2) = 2 + 2 - 3 = 1$.
- The intersection is the line spanned by $(1,1,1,1)$.

Example 3. In $\mathbb{R}^{60}$, $W_1 = \{x_i = 0 \text{ for } 3 \mid i\}$, $W_2 = \{x_i = 0 \text{ for } 4 \mid i\}$. Find both dimensions.

Solution

- $\dim W_1 = 40$, $\dim W_2 = 45$.
- $W_1 \cap W_2$ kills the $20 + 15 - 5 = 30$ indices divisible by 3 or 4: dimension 30.
- $\dim(W_1 + W_2) = 40 + 45 - 30 = 55$: only the 5 multiples of 12 are killed.

Example 4. For which k is $\operatorname{span}\{(1,k,1)\}$ a complement of $U = \operatorname{span}\{(1,1,0), (0,1,1)\}$ in $\mathbb{R}^3$?

Solution

- $U = \{(a, a+b, b)\}$, the plane $y = x + z$.
- $(1,k,1) \in U \iff k = 2$.
- So the line is a complement exactly when $k \neq 2$.

Example 5. Split $A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 3 & 1 \\ 2 & 5 & 1 \end{pmatrix}$ into symmetric and skew-symmetric parts.

Solution

- $S = \frac{1}{2}(A + A^T) = \begin{pmatrix} 1 & 3 & 1 \\ 3 & 3 & 3 \\ 1 & 3 & 1 \end{pmatrix}$.
- $K = \frac{1}{2}(A - A^T) = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & -2 \\ 1 & 2 & 0 \end{pmatrix}$.

Example 6. When is $M_n(\mathbb{F}_p)$ the direct sum of the trace-zero and the scalar matrices?

Solution

- The dimensions are $n^2 - 1$ and 1, so the sum is direct iff the intersection is 0.
- $\operatorname{tr}(cI) = nc$ vanishes for all c iff $p \mid n$.
- So the decomposition holds iff $p \nmid n$; over $\mathbb{R}$ it always holds.

Example 7. In $P_6(\mathbb{R})$, let W_1 be the even polynomials and $W_2 = \{p : p(0) = 0\}$. Find $\dim(W_1 \cap W_2)$ and $W_1 + W_2$.

Solution

- $W_1 = \operatorname{span}\{1, x^2, x^4, x^6\}$, $\dim W_2 = 6$.
- $W_1 \cap W_2 = \operatorname{span}\{x^2, x^4, x^6\}$: dimension 3.
- $\dim(W_1 + W_2) = 4 + 6 - 3 = 7$, so $W_1 + W_2 = P_6$ (the sum is not direct).